

Two-Month Plan: Benchmarking Temporal QA with LLMs

on Machine Prognosis/Diagnosis Data

MTech Research Task — Foundation for Neurosymbolic Temporal Question Answering

Goal of the task: Build a rigorous empirical baseline — “how well do off-the-shelf LLMs (zero-shot, few-shot, RAG) answer temporal questions about machine health, faults, and remaining life?” — that will later be used to show the *gain* a neurosymbolic temporal QA system provides. The deliverable is a benchmark (dataset + evaluation harness + LLM results + failure-mode analysis), not a new model.

Why this scoping works: Temporal QA over general text (news, Wikipedia) is well studied. Temporal QA in the prognosis/diagnosis (PHM — Prognostics & Health Management) domain is not, and it has a distinctive flavor: questions reference sensor-timestamped events, degradation trajectories, fault onset/duration, “before/after/during a maintenance action,” and numeric time-to-failure (RUL) — i.e., quantitative + symbolic temporal reasoning over structured/semi-structured data, not just narrative time. That combination is exactly the gap a neurosymbolic system should close, so it needs to be exposed clearly by the benchmark.

1. Reading List

Organize into three pillars. The student should read Pillar A fully (foundations), skim Pillar B for method ideas, and mine Pillar C for domain data/question templates. Reproduce items marked [REPRODUCE].

Pillar A — Temporal QA Benchmarks & LLM Temporal Reasoning (foundations, general domain)

- **Chen, Wang, Wang (2021):** TimeQA: A Dataset for Answering Time-Sensitive Questions (NeurIPS D&B). Defines time-scoped factual QA; the template most PHM temporal questions will resemble (“what was the fault state at time T / between T1–T2”).
- **Saxena, Chakrabarti, Talukdar (2021):** Question Answering over Temporal Knowledge Graphs (CronQuestions/CRONKGQA, ACL). Establishes the KG-grounded temporal QA setting and operator taxonomy (before/after/first/last/simultaneous).
- **Tan, Ng, Bing (2023) — [REPRODUCE]:** Towards Benchmarking and Improving the Temporal Reasoning Capability of LLMs (ACL, TempReason). Cleanest existing harness for measuring pure LLM temporal reasoning across explicit/implicit/relative time; easiest to adapt to a new domain.
- **Wang & Zhao (2023):** TRAM: Benchmarking Temporal Reasoning for LLMs. Broad taxonomy of temporal reasoning types (ordering, duration, frequency, typical time) — useful as the question-type schema for the PHM benchmark.
- **Chu et al. (2023/2024):** TimeBench: A Comprehensive Evaluation of Temporal Reasoning Abilities in LLMs (ACL 2024). Another taxonomy + evaluation protocol to borrow from.
- **Su et al. (2024):** Temporal Knowledge Graph Question Answering: A Survey (arXiv 2406.14191). Single best survey to get oriented on TKGQA methods, datasets, metrics.
- **Li, Cheng, Tan, Ng, Joty, Bing (2023):** Unlocking Temporal QA for LLMs Using Code Execution. Early neurosymbolic-flavored idea (LLM writes code over temporal facts instead of reasoning purely in text) — directly relevant as a baseline and as inspiration for the follow-on neurosymbolic project.
- **Xiong, Payani, Kompella, Fekri (2024):** Large Language Models Can Learn Temporal Reasoning (ACL). Good for understanding limits/fine-tuning approaches, for later comparison.

Pillar B — Neurosymbolic Temporal Reasoning (method ideas, read selectively)

- **Liang, Zeng, Zhao, Zhao (2025):** NeSTR: A Neuro-Symbolic Abductive Framework for Temporal Reasoning in LLMs. Closest in spirit to what the lab is eventually building; read closely for its evaluation protocol (this is what the benchmark should be able to plug into later).
- **Wang, Sun, Luo, Wei, Hu, Liew, Pan, Yin (2024):** LLM-DA: Large Language Models-guided Dynamic Adaptation for Temporal Knowledge Graph Reasoning. Rule-extraction + LLM hybrid — good concrete example of “neural extracts symbolic rules, symbolic engine reasons.”
- **Tan, Wang, Liu, Xu, Yuan, Zhu, Zhang (2025/2026):** MemoTime: Memory-Augmented Temporal Knowledge Graph Enhanced LLM Reasoning. Shows how to decompose a temporal question into a “Tree of Time” of sub-operators — a good template for the PHM question decomposition too.
- **Optional survey:** Neurosymbolic AI for Reasoning Over Knowledge Graphs: A Survey — skim for the general design space (neural-then-symbolic vs. symbolic-then-neural vs. tightly coupled) so the student can classify where the eventual thesis system sits.

Pillar C — Machine Prognosis/Diagnosis Domain (data sources, question construction)

- **NASA C-MAPSS / N-CMAPSS — [REPRODUCE]:** Turbofan engine degradation datasets (Saxena et al. 2008; Chao, Kulkarni, Goebel, Fink 2021, “Aircraft Engine Run-to-Failure Dataset under Real Flight Conditions,” NASA Prognostics Data Repository). Run a standard RUL baseline (LSTM/CNN) first, purely to understand the sensor/fault-mode/time structure you’ll be writing questions about.
- **CWRU Bearing Data Center:** Classic fault-diagnosis (vibration) benchmark with labeled fault types, sizes, and onset — a second, smaller domain to diversify question sources beyond turbofans.
- **GitHub — CHAOZHAO-1/LLM-based-PHM:** Curated, continuously updated list of “LLM-based fault diagnosis and prognosis” papers. Use as a live index; pull in 3–4 most relevant entries, e.g.:
- – **FaultGPT:** Industrial Fault Diagnosis Question Answering System by Vision-Language Models — closest existing “PHM QA system,” good baseline/comparison and source of QA-pair construction ideas.
- – **FDRKG-LLM — [REPRODUCE-lite]:** A Knowledge-Graph Enhanced LLM-Based Fault Diagnostic Reasoning and Maintenance Decision Support Pipeline towards Industry 5.0. KG + LLM pipeline for fault diagnosis; reproduce the retrieval+prompting pipeline on your own subset even if the exact data isn’t available.
- – **CNCLU dataset:** Intelligent Fault Diagnosis for CNC through Integration of LLMs and Domain Knowledge Graphs (200 Qs). Small, directly usable multiple-choice fault-QA dataset and evaluation format to imitate.
- – **Event-KG fault diagnosis (IEEE TIM 2025):** The Interpretable Reasoning and Intelligent Decision-Making Based on Event Knowledge Graph with LLMs in Fault Diagnosis Scenarios. Event-KG representation of fault timelines, useful for timestamping fault “events” rather than raw sensor readings.
- **Multi-Agent Industrial QA Distillation:** Knowledge Graph-Guided Multi-Agent Distillation for Reliable Industrial QA (with datasets) — ideas on industrial QA dataset construction/validation methodology (multi-agent generation + human/LLM verification loop), needed for building the question set.

Reading pace: ~1 paper/day deep-read for Pillar A+B core (8 papers ≈ 8 days), Pillar C surveyed in parallel while building the dataset (weeks 3–5).

2. Two-Month Plan (8 Weeks)

Week 1 — Orientation & Scoping

- Read Pillar A papers 1–2, 6 (TimeQA, CRONKGQA, TKGQA survey) and Pillar C item on the LLM-based-PHM repo end to end.
- Decide the domain scope: turbofan engine RUL/degradation (C-MAPSS/N-CMAPSS) as primary; CWRU bearings as secondary diversification.
- Write a one-page scoping note: which machine system, which fault modes, what “temporal question” means concretely in this domain (see Section 3).
- Set up compute/repo: Python env, LLM API access (at least 2 model families for comparison), experiment tracking (W&B or CSV logging).

Deliverable: Scoping note + repo skeleton + reading notes for Week 1 papers.

Week 2 — Reproduce a General-Domain Temporal QA Baseline

- [REPRODUCE] TempReason (Tan et al. 2023) harness end-to-end on a small open model and one closed model: get their metric numbers roughly in the right ballpark. This validates the evaluation pipeline before touching PHM data.
- Read Pillar A papers 3–5, 7 in full.

Deliverable: Working eval harness (prompt templates, scoring, logging) validated against a published baseline; short write-up comparing reproduced numbers to the paper's.

Week 3 — Domain Data Ingestion & Event Extraction

- [REPRODUCE] Train/run a standard RUL baseline (simple LSTM or CNN baseline from a public C-MAPSS repo) — goal is not SOTA accuracy, it's to deeply understand the data: run-to-failure trajectories, fault-onset cycle, operating condition regime, sensor drift as timestamped events.
- Convert raw sensor trajectories into a timestamped event log per unit (engine/bearing): degradation-onset time, fault-mode label, condition changes, failure time, cycle index ↔ “timestamp.” This event log is the backbone for question generation.
- Read Pillar C items on C-MAPSS, CWRU, and the CNCLU/FDRKG-LLM papers.

Deliverable: Cleaned event-log representation for ≥2 sub-datasets (e.g., FD001 + FD003 of C-MAPSS, plus CWRU) with a documented schema.

Week 4 — Question Taxonomy & Dataset Construction (Part 1)

- Finalize the PHM-temporal question taxonomy (extend Section 3), covering: point-in-time state queries, before/after/during queries, duration/interval queries, ordering across units or fault events, aggregation, and RUL/forecast queries.
- Build a template + LLM-assisted generation pipeline (semi-automatic, following the multi-agent QA-generation idea): templates instantiated over the event log, LLM used to paraphrase into natural questions, human/rule-based verification against ground truth from the structured log.
- Read NeSTR and MemoTime for how they structure operator-decomposed temporal questions — mirror that structure so the benchmark is neurosymbolic-reasoning-ready later.

Deliverable: Draft question taxonomy (with counts/targets per type) + generation pipeline producing first ~150–200 QA pairs with verified gold answers.

Week 5 — Dataset Construction (Part 2) & Quality Control

- Scale question generation to target size (aim for 800–1500 QA pairs total across question types and both sub-domains).

- Quality control: sample ~10% for human verification, check template diversity, check no answer leakage/trivial shortcuts (e.g., all “yes” answers under one template).
- Split into dev/test; hold out a small “hard” subset (multi-hop / long-horizon) separately.

Deliverable: Finalized PHM-Temporal-QA dataset v1 (data card documenting construction method, taxonomy, size, splits, known limitations).

Week 6 — LLM Benchmarking Runs

- Run the eval harness from Week 2 on the new dataset across conditions: zero-shot, few-shot, and retrieval-augmented (event log / KG context, similar to FDRKG-LLM / CNC-KG pipelines).
- [REPRODUCE-lite] one existing PHM-QA pipeline (e.g., a simplified version of FDRKG-LLM's KG-retrieval-then-generate approach) as an external baseline, not just your own prompting.
- Cover ≥ 2 –3 LLMs of different scale/family for fair comparison.

Deliverable: Results table (accuracy/F1 or exact-match by question type $\times$ model $\times$ prompting condition).

Week 7 — Error Analysis & Failure Taxonomy

- Systematically categorize failures: arithmetic/duration errors, wrong event ordering, hallucinated fault events, unit confusion, context-window/retrieval misses, “implicit time” (e.g., “recently,” “just before failure”) misunderstanding.
- Tie each failure category back to what a neurosymbolic component (temporal operators, event KG grounding, rule-based interval arithmetic) would plausibly fix — the key bridge artifact for the next stage of the lab's research.

Deliverable: Failure-mode report with quantified frequency per category, and a “opportunities for neurosymbolic methods” section mapping failures $\rightarrow$ candidate symbolic components (interval logic, Allen's temporal relations, RUL arithmetic checker, KG-grounded retrieval).

Week 8 — Consolidation, Writing, and Handoff

- Finalize dataset (v1 release), eval harness, and results into a clean repo with README.
- Write it up as a short benchmark paper/technical report (Introduction, Related Work drawing on Pillars A–C, Dataset construction, Experimental setup, Results, Error analysis, Discussion of neurosymbolic opportunities).
- Prepare a slide deck / presentation for the lab summarizing findings and proposing 2–3 concrete neurosymbolic follow-on directions (e.g., “add Allen's interval algebra as a verifier,” “ground questions in an explicit temporal KG and retrieve via structured queries instead of free text”).

Deliverable: Final report + repo + presentation.

3. Starter Question Taxonomy for PHM Temporal QA (to refine in Week 4)

Type	Example
Point-in-time state	What was the operating condition of engine unit 24 at cycle 130?
Before/after	Did the HPT fault appear before or after the LPT fault in unit 7?
Duration/interval	How many cycles elapsed between fault onset and failure for unit 12?
Ordering (cross-unit)	Which of units 3, 9, 15 failed first?
Aggregation over time window	How many units had failed by cycle 150 under condition regime 2?
Forecast/RUL	Given sensor readings up to cycle 90, how many cycles remain before failure?
Counterfactual/relative	Was the degradation rate faster in the second half of the unit's life than

Type	Example
	the first?
Implicit time expression	What fault mode was present just before the engine was taken out of service?

4. Notes for the Advisor / Scope Checks

- Keep the deliverable a benchmark + baseline results, not a new modeling method — that's the next student's job (the neurosymbolic system). Resist scope creep into building the neurosymbolic solution itself.
- Weeks 6–7 are where comparability with the future neurosymbolic system is decided — make sure the dataset format (event logs, KG-able facts, gold answers) is something a symbolic reasoner could also consume, not just free text, or the “foundation for comparison” goal fails.
- If time is tight, the two safest cuts are: reduce to one domain (C-MAPSS only, drop CWRU) and reduce the [REPRODUCE-lite] external baseline to a citation + qualitative comparison instead of a full re-implementation.